

HSRP First Hop Redundancy

Dual-Group Gateway Redundancy with Interface Tracking

01 Overview

The problem

Every host on a subnet is configured with a single default gateway address. If the router holding that address fails, every device behind it loses connectivity beyond its local segment — even when a second, perfectly healthy router sits on the same wire. The hosts have no mechanism to discover it and no way to switch.

Adding a second router does not by itself create redundancy. Without a protocol to share the gateway address between them, the backup router is invisible to the hosts and forwards nothing.

First Hop Redundancy Protocols

An FHRP places a **virtual IP address** on the segment that is shared between two or more routers. Hosts use the virtual IP as their default gateway. The routers elect an Active router among themselves; if it fails, a Standby router assumes the same virtual IP and virtual MAC address. Hosts never change configuration and never learn that anything happened.

Protocol	Standard	Behaviour
HSRP	Cisco proprietary	Active/Standby. One router forwards; the others idle until failure.
VRRP	RFC 5798 (open)	Functionally similar to HSRP. Required in multi-vendor environments.
GLBP	Cisco proprietary	Active/Active. Distributes hosts across multiple forwarders in one group.

This lab implements HSRP version 2.

Objectives

- Provide a redundant default gateway for two separate subnets using a single pair of routers.
- Configure one HSRP group per subnet, with deterministic Active/Standby role assignment.
- Link the two groups with interface tracking so that a failure on either side of a router relinquishes both roles.
- Verify failover empirically under continuous traffic and record the observed convergence.
- Document every fault encountered, its root cause, and the design rule it produced.

02 Topology and Addressing

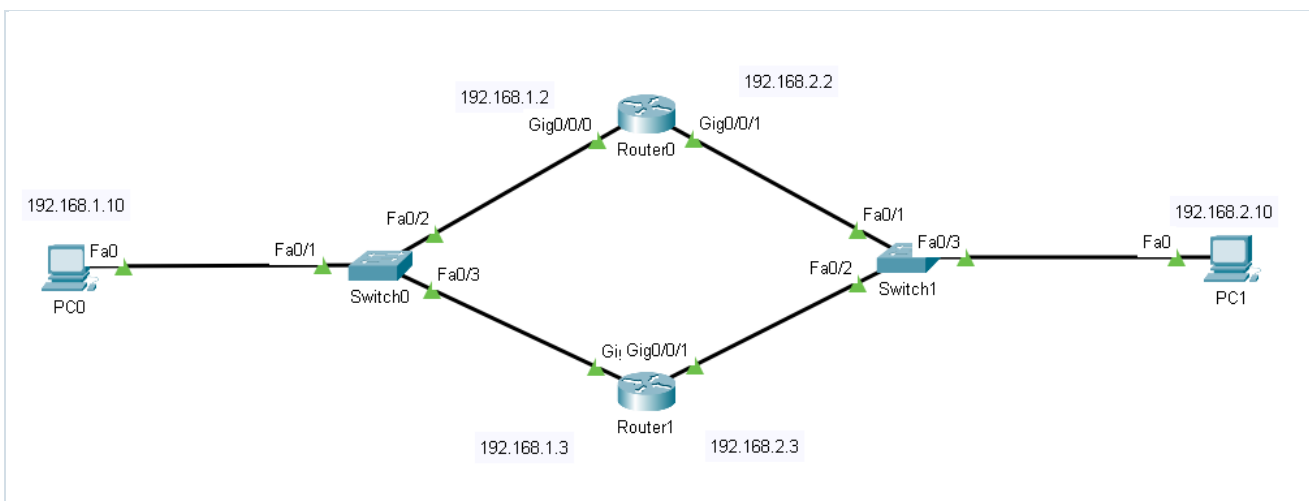


Figure 1 — Lab topology. Both routers hold an interface in each subnet.

Two routers sit between two switches, giving each router a physical leg in **both** subnets. This is the defining structural property of the design: because every subnet is directly connected on both routers, **no routing protocol is required** — each router can already reach both LANs on its own.

It also means **two independent HSRP groups** are needed, one per subnet. HSRP operates per-interface and per-group, not per-router.

Addressing plan

Device	LAN1 — 192.168.1.0/24	LAN2 — 192.168.2.0/24
Router0	G0/0/0 — 192.168.1.2	G0/0/1 — 192.168.2.2
Router1	G0/0/0 — 192.168.1.3	G0/0/1 — 192.168.2.3
HSRP Virtual IP	192.168.1.1 (Group 1)	192.168.2.1 (Group 2)
PC0	192.168.1.10	—
PC1	—	192.168.2.10

CRITICAL

Host default gateways must point at the **virtual IPs** — never at a router's real interface address. Pointing a host at a real address defeats the entire mechanism: the host will keep sending to a dead router indefinitely.

The switches require no configuration. All ports remain access ports in the default VLAN 1.

03 Configuration

Router0 — Active for both groups

```
enable
configure terminal
hostname Router0
!
interface GigabitEthernet0/0/0
description LAN1 - HSRP Group 1
ip address 192.168.1.2 255.255.255.0
standby version 2
standby 1 ip 192.168.1.1
standby 1 priority 110
standby 1 preempt
standby 1 track GigabitEthernet0/0/1 20
no shutdown
!
interface GigabitEthernet0/0/1
description LAN2 - HSRP Group 2
ip address 192.168.2.2 255.255.255.0
standby version 2
standby 2 ip 192.168.2.1
standby 2 priority 110
standby 2 preempt
standby 2 track GigabitEthernet0/0/0 20
no shutdown
!
end
write memory
```

Router1 — Standby for both groups

```
enable
configure terminal
hostname Router1
!
interface GigabitEthernet0/0/0
description LAN1 - HSRP Group 1
ip address 192.168.1.3 255.255.255.0
standby version 2
standby 1 ip 192.168.1.1
standby 1 preempt
no shutdown
!
interface GigabitEthernet0/0/1
description LAN2 - HSRP Group 2
ip address 192.168.2.3 255.255.255.0
standby version 2
standby 2 ip 192.168.2.1
standby 2 preempt
no shutdown
!
end
write memory
```

NOTE

Router1's priority is deliberately left at the default of 100. However `preempt` is configured here even though it is the lower-priority router — without it, Router1 will never take over when tracking lowers Router0's priority. The tracking design depends on `preempt` being present on **both** routers.

Command reference

Command	Purpose
<code>standby version 2</code>	Selects HSRPv2. Required for group numbers above 255 and for IPv6. Must match on both routers — v1 and v2 use different multicast addresses and will not peer.
<code>standby N ip <VIP></code>	Defines the virtual IP address that hosts will use as their default gateway.
<code>standby N priority <1-255></code>	Election weight. Highest value wins the Active role. Default is 100.
<code>standby N preempt</code>	Allows a router to take the Active role when it holds the higher priority. Without it, whichever router became Active first keeps the role indefinitely.
<code>standby N track <int> <dec></code>	Monitors a different interface. If that interface goes down, this group's priority is reduced by the decrement value.
<code>standby N timers msec H msec D</code>	Adjusts hello and hold timers. Defaults are 3 s hello / 10 s hold. Must match on both routers.

Why interface tracking matters

Without tracking, HSRP reacts only to the failure of the interface the group is configured on. It has no awareness of the rest of the router. Consider Router0 losing its LAN2 interface while its LAN1 interface stays up:

- Group 1 on LAN1 sees nothing wrong. Router0 remains Active.
- PC0 continues sending all off-subnet traffic to Router0.
- Router0 has no path to LAN2 — its only route was the connected interface that just went down.
- **Traffic is silently discarded. Router1 is healthy and idle throughout.**

This is a **black hole**: the control plane reports a healthy Active router while the data plane drops every packet. It is worse than a total router failure, because a total failure would at least trigger failover.

Tracking closes the gap by linking the two groups. A failure on either interface lowers the priority on both, handing the Active role to Router1 for the router *as a whole* rather than one interface at a time. This fault was reproduced during the build and is documented as Finding 3.

04 Verification

`show standby brief` is the primary verification command. It reports one line per group and exposes every common misconfiguration at a glance.

```
Router#show standby brief
                P indicates configured to preempt.
                |
Interface    Grp  Pri P State    Active        Standby        Virtual IP
Gig0/0/0     1   110 P Active    local         192.168.1.3    192.168.1.1
Gig0/0/1     2   110 P Active    local         192.168.2.3    192.168.2.1
```

Figure 2 — Router0: Active for both groups, priority 110, preempt enabled.

```

Router#show standby brief
                P indicates configured to preempt.
                |
Interface    Grp   Pri P State      Active        Standby        Virtual IP
Gig0/0/0     1     100 Standby  192.168.1.2   local         192.168.1.1
Gig0/0/1     2     100 P Standby  192.168.2.2   local         192.168.2.1

```

Figure 3 — Router1: Standby for both groups, default priority 100.

What a healthy pair looks like

- One router reports Active and the other Standby, for each group.
- The Virtual IP column matches on both routers.
- The Standby column is populated with the peer's real IP address — **never** the value unknown.
- The **P** flag is present on every line, confirming preempt is configured.

DIAGNOSTIC VALUE

The **Standby unknown** field is the single fastest indicator of a peering failure, and it was the symptom that exposed two of the three faults found in this lab. If a router cannot name its peer, the two devices are not exchanging hellos — regardless of what the rest of the configuration looks like.

05 Failover Testing

Method

A continuous ping is started from PC0 to PC1, and only then is the Active router's interface administratively shut. Order matters — see the measurement note below.

```
PC0> ping -t 192.168.2.10

Router0(config)# interface gig0/0/0
Router0(config-if)# shutdown
```

Result

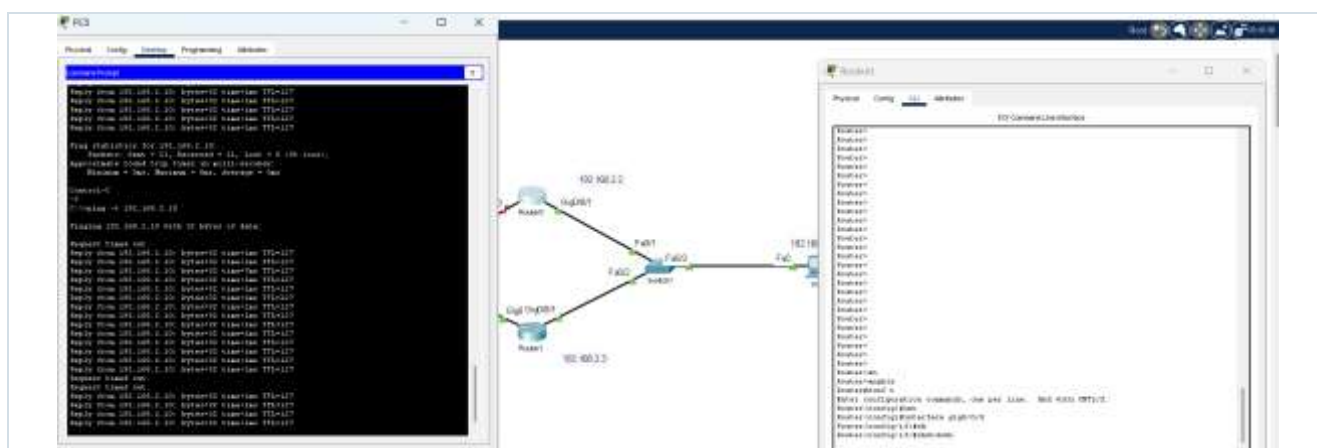


Figure 4 — Continuous ping across the failover event. Two packets lost, then service restored via Router1.

Two packets were dropped and traffic resumed automatically through Router1. No host reconfiguration took place; PC0 continued sending to 192.168.2.1 throughout.

The gap corresponds to the HSRP hold time — by default 10 seconds, derived from three missed 3-second hellos. Sub-second convergence is available by tightening the timers on both routers:

```
standby 1 timers msec 200 msec 750
```

Hello 200 ms, hold 750 ms. Timer values must match on both routers.

Measurement note

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127
Reply from 192.168.2.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 5 — A fixed-count ping issued after the shutdown shows 25% loss.

INTERPRETATION

This is a **measurement artifact, not a failure**. A four-packet ping issued immediately after the event falls inside the dead interval, so the first probe times out before the Standby has assumed the role. Testing an FHRP with a fixed-count ping started after the failure will always understate its performance.

The ping must already be running before the failure is induced, as in Figure 4.

06 Troubleshooting Log

Three faults were found and corrected during the build. All three share a characteristic worth stating up front:

KEY OBSERVATION

None of the three faults produced a syntax error, a rejected command, or a log message.

Each configuration was accepted in full by the CLI and simply did not behave as intended. This is the defining difficulty of FHRP troubleshooting — the device reports success while the redundancy silently does not work. Verification must be behavioural, not visual.

Finding 1 — HSRP version mismatch

Symptom — Both routers reported Active for the same group, each showing Standby unknown.

```
Router#show standby brief
                P indicates configured to preempt.
                |
Interface    Grp  Pri  P State      Active        Standby      Virtual IP
Gig0/0/0     1   100  Standby  192.168.1.2   local        192.168.1.1
Gig0/0/1     2   100  P Active   local         unknown      192.168.2.1
Router#
%HSRP-6-STATECHANGE: GigabitEthernet0/0/0 Grp 1 state Standby -> Active
```

Figure 6 — Both routers claiming the Active role for their own groups, with no peer detected.

Cause — One router was running HSRPv1 and the other HSRPv2. The two versions use different multicast destination addresses: **224.0.0.2** for v1 and **224.0.0.102** for v2. Neither router ever received the other's hellos, so each concluded it was alone on the segment and elected itself Active.

Impact — Split-brain. Two routers answering ARP for the same virtual IP with different virtual MAC addresses, causing host ARP caches to flap and producing intermittent, hard-to-attribute packet loss.

Fix — `standby version 2` applied consistently on both routers.

COMMAND ORDER

Changing the HSRP version **clears existing group configuration** on that interface. The `standby version 2` line must be entered *before* the group commands, or those commands are silently discarded.

Finding 2 — Group number mismatch

Symptom — Identical to Finding 1: both routers Active, Standby unknown on both.

Cause — Router0 had group 0 configured on G0/0/1 while Router1 had group 2, both advertising virtual IP 192.168.2.1. HSRP peers exchange hellos only within the same group number, so the two routers were invisible to one another despite sharing a subnet and a virtual IP.

Impact — Split-brain again, with an additional subtlety: the virtual MAC is derived from the group number, so group 0 produced 0000.0c9f.f000 and group 2 produced 0000.0c9f.f002 for the same IP address.

Fix — Standardised on group 2 for the 192.168.2.0/24 segment across both routers.

Finding 3 — Tracking decrement too small to trigger failover

Symptom — With Router0's LAN2 interface shut, PC0 could still reach 192.168.1.1 and 192.168.1.3 but could not reach 192.168.2.1 or PC1. Router1 was healthy and could ping both hosts.

Cause — The tracking statements had been entered without a decrement value:

```
standby 1 track GigabitEthernet0/0/1      ← no decrement specified
```

This does not disable tracking. It applies the **default decrement of 10**. With Router0 at priority 110 and Router1 at 100, a tracked failure produced:

Router	Priority before	Priority after failure	Result
Router0	110	110 – 10 = 100	Remains Active
Router1	100	100	Remains Standby

A tie. Preempt fires only on a **strictly higher** priority, so Router1 never took the role. Router0 stayed Active on LAN1 while black-holing everything destined for LAN2 — precisely the failure mode that tracking exists to prevent.

Fix — Specify the decrement explicitly.

```
standby 1 track GigabitEthernet0/0/1 20
```

110 – 20 = 90, strictly below Router1's 100. Preempt fires and failover completes.

DESIGN RULE EXTRACTED

The tracking decrement must be strictly greater than the priority gap between the routers.

A 110 / 100 pair requires a decrement of at least 11. The conventional choice is 20, which leaves clear margin and survives a later priority adjustment without silently breaking failover.

Simulator constraint

Packet Tracer's HSRP parser does not accept the decrement argument on `standby N track` in all builds, returning `invalid command` while accepting the bare interface form. Where the decrement cannot be set, the same outcome is achieved by narrowing the priority gap to below the fixed default decrement of 10:

```
standby 1 priority 105      ! 105 - 10 = 95 < 100 → preempt fires
```

SCOPE OF THIS WORKAROUND

Production configurations should use the explicit `110 / 100 / decrement 20` form. The 105 variant compensates for a simulator limitation and is not a design recommendation.

07 Design Notes

Active/Standby versus load sharing

As built, Router0 is Active for both groups and Router1 forwards no traffic until a failure occurs. Half the available forwarding capacity is idle at all times.

Reversing the priorities on Group 2 — Router1 at 110, Router0 at 100 — makes each router Active for one subnet and Standby for the other. Both devices then carry production traffic, failover behaviour is unchanged, and each router's health is continuously proven by live traffic rather than by hello packets alone. This is the more common real-world deployment and follows directly from HSRP being a *per-group* protocol rather than a per-router one.

Convergence and timer selection

Default timers give roughly 10 seconds of loss. For voice, video, or industrial control traffic this is too slow — many ICS protocols will drop a session well inside that window. Millisecond timers bring convergence under one second, at the cost of additional hello traffic and reduced tolerance for transient link errors. Timers must be configured identically on every router in the group.

Protocol selection

Requirement	Appropriate protocol
All-Cisco environment, Active/Standby acceptable	HSRP
Multi-vendor environment	VRRP (RFC 5798)
Idle standby capacity unacceptable; both routers must forward within one group	GLBP

Relevance to OT and ICS environments

Gateway redundancy is an availability control, and availability is the primary security objective in industrial environments — the inverse of the traditional IT priority order. A default gateway failure at the boundary of a control network can isolate an HMI from its historian, or a supervisory zone from its engineering workstation, with consequences that extend beyond data loss.

FHRP deployment at zone boundaries supports the availability requirements set out in **IEC 62443-3-3** (system requirements, particularly the resource availability family) and complements the segmented conduit architecture defined in **IEC 62443-3-2**. The black-hole scenario documented in Finding 3 is especially relevant here: a redundancy mechanism that reports healthy while silently discarding traffic is more dangerous in a control environment than no redundancy at all, because it defeats the operator's assumption that a failure would be visible.